

Germination analysis

masrir

2026-06-07

```
# Reload both datasets
```

```
data <- read.csv("~/Desktop/UUIDS_DatePalms/MH.csv")
data1 <- read.csv("~/Desktop/UUIDS_DatePalms/ML.csv")
str(data)
```

```
## 'data.frame': 200 obs. of 6 variables:
## $ S. : int 1 2 3 4 5 6 7 8 9 10 ...
## $ UUID : chr "29f6b0cb-8ddf-43ff-a816-221215c80fe3" "291c871b-10d7-44c1-bc55-bcdabae9ee3a" "54..."
## $ Source : chr "Aldka" "Aldka" "Aldka" "Aldka" ...
## $ Cult : chr "Majdool" "Majdool" "Majdool" "Majdool" ...
## $ Quality : chr "H" "H" "H" "H" ...
## $ Germinate: chr "" "05/04/2025" "07/04/2025" "06/04/2025" ...
```

```
str(data1)
```

```
## 'data.frame': 200 obs. of 6 variables:
## $ S. : int 1 2 3 4 5 6 7 8 9 10 ...
## $ UUID : chr "b38fb90e-a852-4e4f-8c03-57a44a770bc4" "97402aca-b79e-4c62-a6c8-c6d588cdb997" "2d..."
## $ Source : chr "Aldka" "Aldka" "Aldka" "Aldka" ...
## $ Cult : chr "Majdool" "Majdool" "Majdool" "Majdool" ...
## $ Quality : chr "L" "L" "L" "L" ...
## $ Germinate: chr "" "" "15/04/2025" "06/04/2025" ...
```

```
# Subset relevant columns from data
```

```
data_sub <- data[, c("S.", "UUID", "Cult", "Quality", "Germinate")]
str(data_sub)
```

```
## 'data.frame': 200 obs. of 5 variables:
## $ S. : int 1 2 3 4 5 6 7 8 9 10 ...
## $ UUID : chr "29f6b0cb-8ddf-43ff-a816-221215c80fe3" "291c871b-10d7-44c1-bc55-bcdabae9ee3a" "54..."
## $ Cult : chr "Majdool" "Majdool" "Majdool" "Majdool" ...
## $ Quality : chr "H" "H" "H" "H" ...
## $ Germinate: chr "" "05/04/2025" "07/04/2025" "06/04/2025" ...
```

```
# Subset relevant columns from data1
```

```
data1_sub <- data1[, c("S.", "UUID", "Cult", "Quality", "Germinate")]
str(data1_sub)
```

```
## 'data.frame': 200 obs. of 5 variables:
```

```
## $ S.      : int  1 2 3 4 5 6 7 8 9 10 ...
## $ UUID    : chr  "b38fb90e-a852-4e4f-8c03-57a44a770bc4" "97402aca-b79e-4c62-a6c8-c6d588cdb997" "2d
## $ Cult    : chr  "Majdool" "Majdool" "Majdool" "Majdool" ...
## $ Quality : chr  "L" "L" "L" "L" ...
## $ Germinate: chr  "" "" "15/04/2025" "06/04/2025" ...
```

```
# Combine
combined_data <- rbind(data_sub, data1_sub)
str(combined_data)
```

```
## 'data.frame':  400 obs. of  5 variables:
## $ S.      : int  1 2 3 4 5 6 7 8 9 10 ...
## $ UUID    : chr  "29f6b0cb-8ddf-43ff-a816-221215c80fe3" "291c871b-10d7-44c1-bc55-bcdabae9ee3a" "54
## $ Cult    : chr  "Majdool" "Majdool" "Majdool" "Majdool" ...
## $ Quality : chr  "H" "H" "H" "H" ...
## $ Germinate: chr  "" "05/04/2025" "07/04/2025" "06/04/2025" ...
```

```
today <- as.Date("2025-06-30")
```

```
# Convert germination date
combined_data$germ_date <- as.Date(combined_data$Germinate, format = "%d/%m/%Y")
# Assign correct planting date (fixed version)
combined_data$planting_date <- as.Date(ifelse(combined_data$Quality== "H",
"2025-02-19",
"2025-02-27"))
# If germinated, time = germination - planting
# If not germinated, time = today - planting (right censored)
combined_data$time <- ifelse(
is.na(combined_data$germ_date),
as.numeric(today- combined_data$planting_date),
as.numeric(combined_data$germ_date- combined_data$planting_date)
)
# Status: 1 = germinated, 0 = still waiting
combined_data$status <- ifelse(is.na(combined_data$germ_date), 0, 1)
```

```
library(survival)
library(survminer)
```

```
## Loading required package: ggplot2
```

```
## Loading required package: ggpubr
```

```
##
## Attaching package: 'survminer'
```

```
## The following object is masked from 'package:survival':
##
##      myeloma
```

```

# Build survival object
surv_object <- Surv(time = combined_data$time, event = combined_data$status)
# Fit survival model
fit <- survfit(surv_object ~ Quality, data = combined_data)
survdifff(surv_object ~ Quality, data = combined_data)

```

```

## Call:
## survdiff(formula = surv_object ~ Quality, data = combined_data)
##
##              N Observed Expected (O-E)^2/E (O-E)^2/V
## Quality=H 200      162      131      7.53      15
## Quality=L 200      120      151      6.50      15
##
##  Chisq= 15  on 1 degrees of freedom, p= 1e-04

```

```

# Plot it
library(survival)
library(survminer)
p1 <- ggsurvplot(
  fit,
  data = combined_data,
  pval = TRUE,
  pval.coord = c(90, 0.2),
  fun = "event",
  xlab = "Days since planting",
  ylab = "Cumulative Germination Probability",
  legend.title = "",
  legend.labs = c("Majdool High", "Low"),
  palette = c("darkgreen", "orange"),
  xlim = c(0, 130),
  ylim = c(0, 1),
  break.time.by = 20,
  ggtheme = theme_minimal() +
    theme(
      legend.text = element_text(size = 18)
    )
)

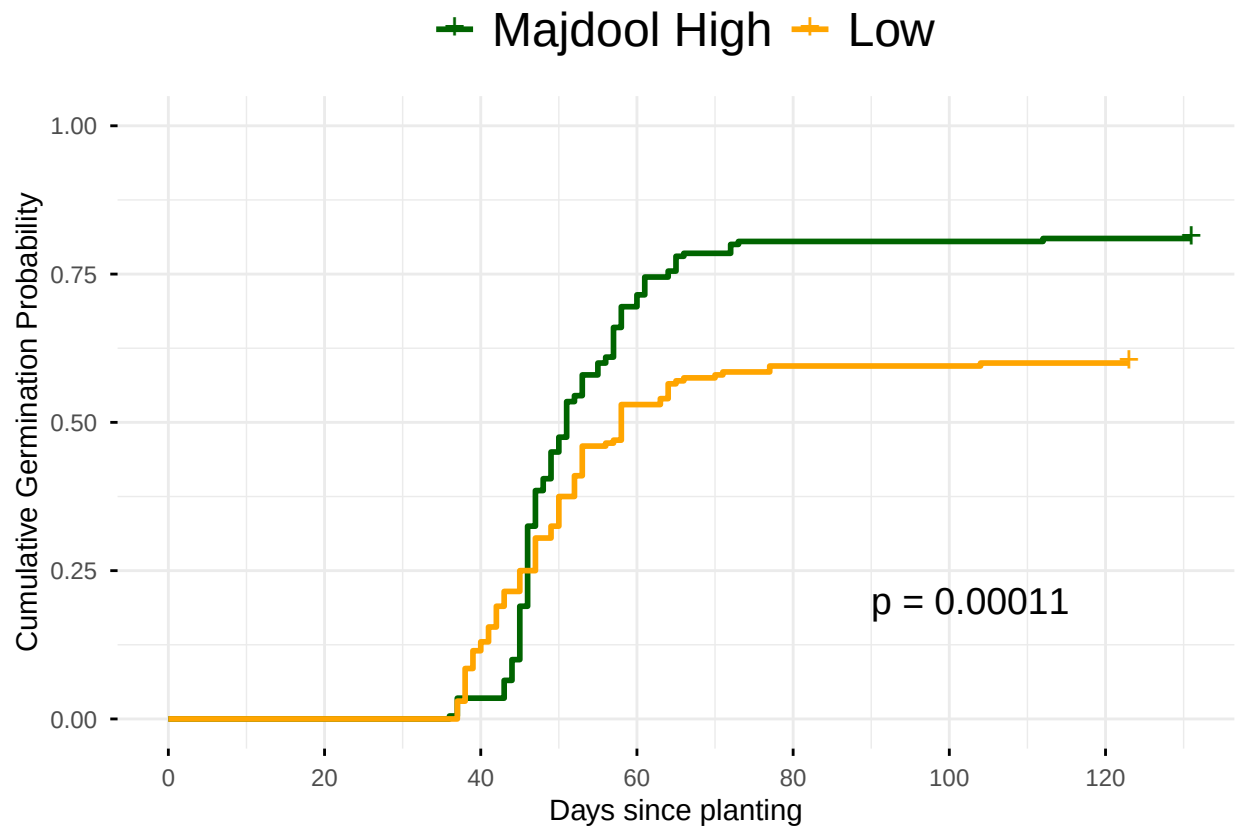
```

```

## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## i The deprecated feature was likely used in the ggpubr package.
## Please report the issue at <https://github.com/kassambara/ggpubr/issues>.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.

```

```
p1
```



```
prop_table <- with(combined_data, tapply(status, Quality, mean, na.rm = TRUE))
prop_table
```

```
##      H      L
## 0.81 0.60
```

```
# Reload both datasets for khudri
data3 <- read.csv("~/Desktop/UUIDS_DatePalms/KH.csv")
data4 <- read.csv("~/Desktop/UUIDS_DatePalms/KL.csv")
str(data3)
```

```
## 'data.frame':   200 obs. of  6 variables:
## $ S.           : int  1 2 3 4 5 6 7 8 9 10 ...
## $ UUID         : chr  "6c6dd6eb-63a7-4561-bfac-ce9057aca483" "db2f0e89-9baf-4eef-b65e-7d6df8322f46" "aa
## $ Source       : chr  "Aldka" "Aldka" "Aldka" "Aldka" ...
## $ Cult         : chr  "Khudri" "Khudri" "Khudri" "Khudri" ...
## $ Quality      : chr  "H" "H" "H" "H" ...
## $ Germinate    : chr  "" "" "" "" ...
```

```
str(data4)
```

```
## 'data.frame':   200 obs. of  6 variables:
## $ S.           : int  1 2 3 4 5 6 7 8 9 10 ...
## $ UUID         : chr  "387afec4-b30d-4dd3-beeb-73a5dc24d8da" "0a9e271a-cc5c-42fc-9eaf-2f5148b87d07" "45
```

```
## $ Source : chr "Aldka" "Aldka" "Aldka" "Aldka" ...
## $ Cult : chr "Khudri" "Khudri" "Khudri" "Khudri" ...
## $ Quality : chr "L" "L" "L" "L" ...
## $ Germinate: chr "" "01/05/2025" "" "21/04/2025" ...
```

```
# Subset relevant columns from data3
```

```
data3_sub <- data3[, c("S.", "UUID", "Cult", "Quality", "Germinate")]
str(data3_sub)
```

```
## 'data.frame': 200 obs. of 5 variables:
## $ S. : int 1 2 3 4 5 6 7 8 9 10 ...
## $ UUID : chr "6c6dd6eb-63a7-4561-bfac-ce9057aca483" "db2f0e89-9baf-4eef-b65e-7d6df8322f46" "aa
## $ Cult : chr "Khudri" "Khudri" "Khudri" "Khudri" ...
## $ Quality : chr "H" "H" "H" "H" ...
## $ Germinate: chr "" "" "" "" ...
```

```
# Subset relevant columns from data4
```

```
data4_sub <- data4[, c("S.", "UUID", "Cult", "Quality", "Germinate")]
str(data4_sub)
```

```
## 'data.frame': 200 obs. of 5 variables:
## $ S. : int 1 2 3 4 5 6 7 8 9 10 ...
## $ UUID : chr "387afec4-b30d-4dd3-beeb-73a5dc24d8da" "0a9e271a-cc5c-42fc-9eaf-2f5148b87d07" "45
## $ Cult : chr "Khudri" "Khudri" "Khudri" "Khudri" ...
## $ Quality : chr "L" "L" "L" "L" ...
## $ Germinate: chr "" "01/05/2025" "" "21/04/2025" ...
```

```
combined_data34 <- rbind(data3_sub, data4_sub)
str(combined_data34)
```

```
## 'data.frame': 400 obs. of 5 variables:
## $ S. : int 1 2 3 4 5 6 7 8 9 10 ...
## $ UUID : chr "6c6dd6eb-63a7-4561-bfac-ce9057aca483" "db2f0e89-9baf-4eef-b65e-7d6df8322f46" "aa
## $ Cult : chr "Khudri" "Khudri" "Khudri" "Khudri" ...
## $ Quality : chr "H" "H" "H" "H" ...
## $ Germinate: chr "" "" "" "" ...
```

```
today <- as.Date("2025-06-30") # or Sys.Date()
```

```
# Convert germination date
```

```
combined_data34$germ_date <- as.Date(combined_data34$Germinate,
format = "%d/%m/%Y")
```

```
# Assign correct planting date (fixed version)
```

```
combined_data34$planting_date <- as.Date(ifelse(combined_data34$Quality == "H",
"2025-03-05",
"2025-03-12"))
```

```
# If germinated, time = germination - planting
```

```
# If not germinated, time = today - planting (right censored)
```

```
combined_data34$time <- ifelse(
is.na(combined_data34$germ_date),
as.numeric(today - combined_data34$planting_date),
```

```

as.numeric(combined_data34$germ_date- combined_data34$planting_date)
)
# Status: 1 = germinated, 0 = still waiting
combined_data34$status <- ifelse(is.na(combined_data34$germ_date), 0, 1)
library(survival)
library(survminer)
# Build survival object
surv_object <- Surv(time = combined_data34$time, event = combined_data34$status)
# Fit survival model
fit <- survfit(surv_object ~ Quality, data = combined_data34)
survdifff(surv_object ~ Quality, data = combined_data34)

```

```

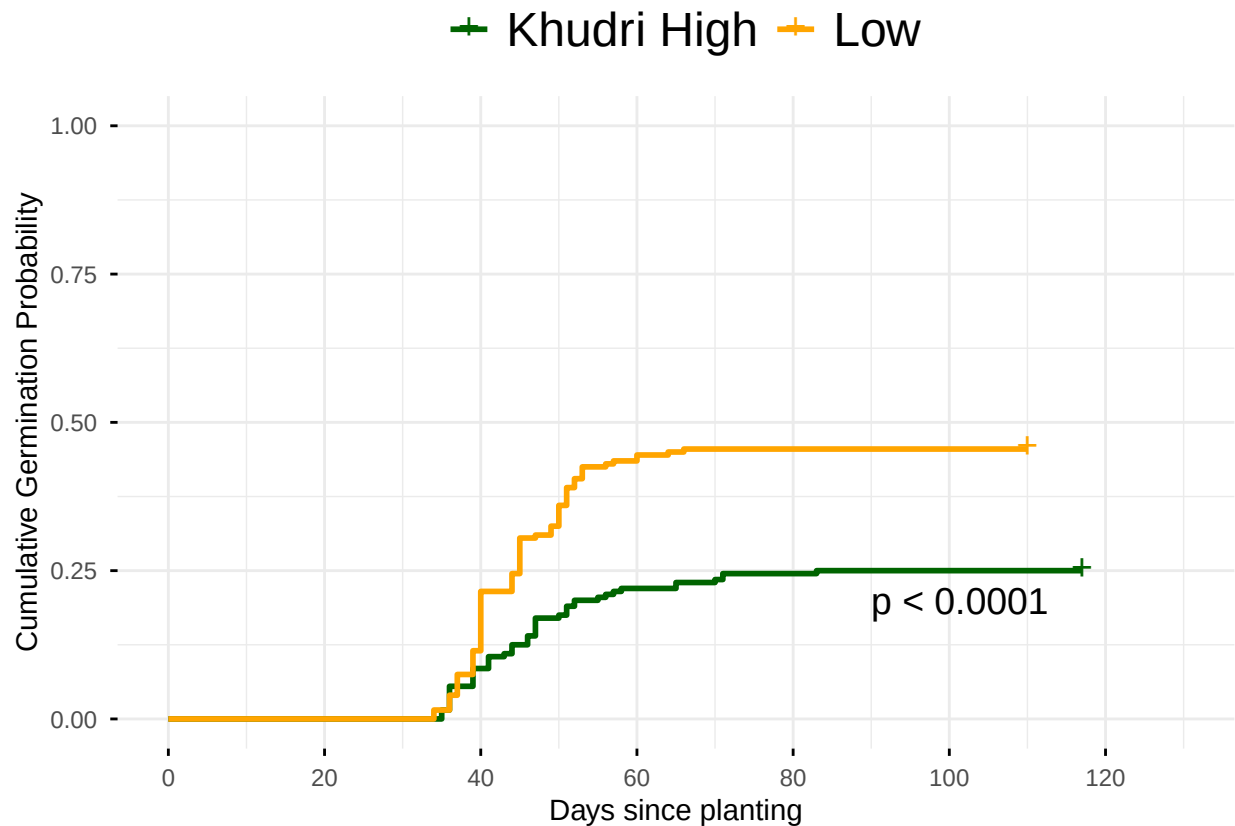
## Call:
## survdiff(formula = surv_object ~ Quality, data = combined_data34)
##
##              N Observed Expected (O-E)^2/E (O-E)^2/V
## Quality=H 200      50    75.3      8.48      18.7
## Quality=L 200      91    65.7      9.71      18.7
##
##  Chisq= 18.7  on 1 degrees of freedom, p= 2e-05

```

```

# Plot it
p2 <- ggsurvplot(
  fit,
  data = combined_data34,
  pval = TRUE,
  pval.coord = c(90, 0.2),
  fun = "event",
  xlab = "Days since planting",
  ylab = "Cumulative Germination Probability",
  legend.title = "",
  legend.labs = c("Khudri High", "Low"),
  palette = c("darkgreen", "orange"),
  xlim = c(0,130),
  ylim = c(0,1),
  break.time.by = 20,
  ggtheme = theme_minimal() +
    theme(
      legend.text = element_text(size = 18)
    )
)
p2

```



```
prop_table <- with(combined_data34, tapply(status, Quality, mean, na.rm = TRUE))
prop_table
```

```
##      H      L
## 0.250 0.455
```

```
# Reload both datasets for saqi
data5 <- read.csv("~/Desktop/UUIDS_DatePalms/SH.csv")
data6 <- read.csv("~/Desktop/UUIDS_DatePalms/SL.csv")
str(data5)
```

```
## 'data.frame':   200 obs. of  6 variables:
## $ S.           : int  1 2 3 4 5 6 7 8 9 10 ...
## $ UUID         : chr  "ab4dc4a5-8980-4ac1-9124-2979ba49143c" "eaf344e8-27e2-4749-8326-810f174726c3" "12..."
## $ Source       : chr  "Aldka" "Aldka" "Aldka" "Aldka" ...
## $ Cult         : chr  "Saqi" "Saqi" "Saqi" "Saqi" ...
## $ Quality      : chr  "H" "H" "H" "H" ...
## $ Germinate    : chr  "" "" "" "" ...
```

```
str(data6)
```

```
## 'data.frame':   200 obs. of  6 variables:
## $ S.           : int  1 2 3 4 5 6 7 8 9 10 ...
## $ UUID         : chr  "e2f98b11-6e3b-4a2b-8f1a-d65b0c64f642" "d9f87791-e38d-4d67-bf50-d0e666df9c1a" "41..."
```

```
## $ Source : chr "Aldka" "Aldka" "Aldka" "Aldka" ...
## $ Cult : chr "Saqi" "Saqi" "Saqi" "Saqi" ...
## $ Quality : chr "L" "L" "L" "L" ...
## $ Germinate: chr "26/04/2025" "26/04/2025" "26/04/2025" "26/04/2025" ...
```

```
# Subset relevant columns from data5
```

```
data5_sub <- data5[, c("S.", "UUID", "Cult", "Quality", "Germinate")]
str(data5_sub)
```

```
## 'data.frame': 200 obs. of 5 variables:
## $ S. : int 1 2 3 4 5 6 7 8 9 10 ...
## $ UUID : chr "ab4dc4a5-8980-4ac1-9124-2979ba49143c" "eaf344e8-27e2-4749-8326-810f174726c3" "12..."
## $ Cult : chr "Saqi" "Saqi" "Saqi" "Saqi" ...
## $ Quality : chr "H" "H" "H" "H" ...
## $ Germinate: chr "" "" "" "" ...
```

```
# Subset relevant columns from data6
```

```
data6_sub <- data6[, c("S.", "UUID", "Cult", "Quality", "Germinate")]
str(data6_sub)
```

```
## 'data.frame': 200 obs. of 5 variables:
## $ S. : int 1 2 3 4 5 6 7 8 9 10 ...
## $ UUID : chr "e2f98b11-6e3b-4a2b-8f1a-d65b0c64f642" "d9f87791-e38d-4d67-bf50-d0e666df9c1a" "41..."
## $ Cult : chr "Saqi" "Saqi" "Saqi" "Saqi" ...
## $ Quality : chr "L" "L" "L" "L" ...
## $ Germinate: chr "26/04/2025" "26/04/2025" "26/04/2025" "26/04/2025" ...
```

```
# Combine
```

```
combined_data56 <- rbind(data5_sub, data6_sub)
str(combined_data56)
```

```
## 'data.frame': 400 obs. of 5 variables:
## $ S. : int 1 2 3 4 5 6 7 8 9 10 ...
## $ UUID : chr "ab4dc4a5-8980-4ac1-9124-2979ba49143c" "eaf344e8-27e2-4749-8326-810f174726c3" "12..."
## $ Cult : chr "Saqi" "Saqi" "Saqi" "Saqi" ...
## $ Quality : chr "H" "H" "H" "H" ...
## $ Germinate: chr "" "" "" "" ...
```

```
today <- as.Date("2025-06-30")
```

```
# Convert germination date
```

```
combined_data56$germ_date <- as.Date(combined_data56$Germinate,
format = "%d/%m/%Y")
```

```
# Assign correct planting date (fixed version)
```

```
combined_data56$planting_date <- as.Date(ifelse(combined_data56$Quality == "H",
"2025-03-19",
"2025-03-17"))
```

```
# If germinated, time = germination - planting
```

```
# If not germinated, time = today - planting (right censored)
```

```
combined_data56$time <- ifelse(
is.na(combined_data56$germ_date),
```



```

as.numeric(today- combined_data56$planting_date),
as.numeric(combined_data56$germ_date- combined_data56$planting_date)
)
# Status: 1 = germinated, 0 = still waiting
combined_data56$status <- ifelse(is.na(combined_data56$germ_date), 0, 1)
library(survival)
library(survminer)
# Build survival object
surv_object <- Surv(time = combined_data56$time, event = combined_data56$status)
# Fit survival model
fit <- survfit(surv_object ~ Quality, data = combined_data56)
survdifff(surv_object ~ Quality, data = combined_data56)

```

```

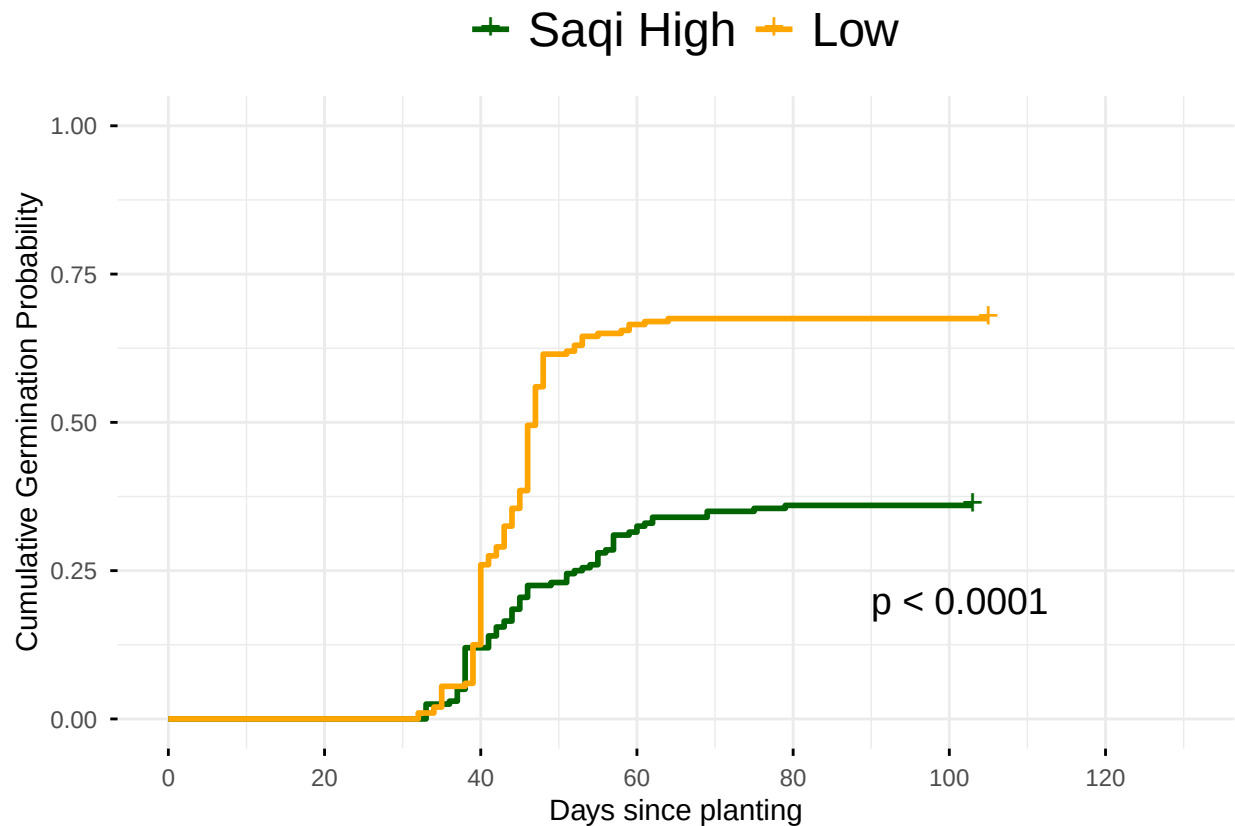
## Call:
## survdiff(formula = surv_object ~ Quality, data = combined_data56)
##
##              N Observed Expected (O-E)^2/E (O-E)^2/V
## Quality=H 200      72   116.6      17.1      41.4
## Quality=L 200     135    90.4      22.0      41.4
##
##  Chisq= 41.4  on 1 degrees of freedom, p= 1e-10

```

```

# Plot it
p3 <- ggsurvplot(
  fit,
  data = combined_data56,
  pval = TRUE,
  pval.coord = c(90, 0.2),
  fun = "event",
  xlab = "Days since planting",
  ylab = "Cumulative Germination Probability",
  legend.title = "",
  legend.labs = c("Saqi High", "Low"),
  palette = c("darkgreen", "orange"),
  xlim = c(0,130),
  ylim = c(0,1),
  break.time.by = 20,
  ggtheme = theme_minimal() +
    theme(
      legend.text = element_text(size = 18) # increase legend label size
    )
)
p3

```



```
prop_table <- with(combined_data56, tapply(status, Quality, mean, na.rm = TRUE))
prop_table
```

```
##      H      L
## 0.360 0.675
```

```
library(survminer)
plots_list <- list(p1, p2, p3)
combined <- arrange_ggsurvplots(
  plots_list,
  ncol = 3,
  nrow = 1,
  print = FALSE, size=20
)
png("~/Desktop/Figure4_2_germination.png",
     width = 30, height = 10, units = "cm", res = 300)
```

```
combined
```

```
library(survival)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
get_rmst <- function(data, cultivar_name){

  surv_obj <- Surv(data$time, data$status)

  fit <- survfit(surv_obj ~ Quality, data = data)

  rmst <- summary(fit)$table[, "rmean"]

  germ_pct <- data %>%
    group_by(Quality) %>%
    summarise(
      Germination_Percent =
        round(mean(status) * 100, 1)
    )

  final <- data.frame(
    Cultivar = cultivar_name,
    Quality = c("H", "L"),
    RMST_Days = round(rmst,1),
    Germination_Percent = germ_pct$Germination_Percent
  )

  return(final)
}

majdool <- get_rmst(combined_data, "Majdool")
khudri <- get_rmst(combined_data34, "Khudri")
saqi <- get_rmst(combined_data56, "Saqi")

final_table <- rbind(majdool, khudri, saqi)

final_table
```

```
##           Cultivar Quality RMST_Days Germination_Percent
## Quality=H   Majdool      H      66.1             81.0
## Quality=L   Majdool      L      82.0             60.0
## Quality=H1   Khudri      H      99.5             25.0
## Quality=L1   Khudri      L      84.0             45.5
## Quality=H2    Saqi      H      84.1             36.0
## Quality=L2    Saqi      L      63.7             67.5
```